CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements — Team Template

Weight 4% · Due Sunday 18 October 2026, 11:59 PM Central · Submit one PDF per team on Canvas

Field	Your entry
Project title	MidMeet
Team number and section	Team 3 3354.005
Team members (name · net ID)	Nicholas Jackson (dal382622), Nandhana Kamalabaskaran (NSK230006), Sameeraa Kandalgaonkar (SXK220350), Alison Kao(ayk230001), Yair Parkhov(yxp240007), Vedanshi Sreeramoju (vxs230103), Tanay Munnangi(tnm230005)
Repository URL	Repository URL
Document version	0.1
Date	10/18/26

0 Revision History and Contribution Record

Both tables below are graded. The contribution record carries the individual 20% of this milestone and must be consistent with your repository history.

Version	Date	Author	Summary of change
0.1	10/18/26	Team 3	Initial baseline for P1

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
Vedanshi Sreeramoju	2.1, 2.5 (A2-A4), 3.4, 3.5	Issue 2
Yair Parkhov	Reviewed and edited 2.1,3.4,3.5	Issue 5
Alison Kao	1.1, 1.2, 1.3, 3.1, 3.2, 3.3	Issue 1
Nandhana Kamalabaskaran	4.1, 4.2, 4.3, 4.4, 4.5, 5	Issue 4
Sameeraa Kandalgaonkar	2.2, 2.3, 2.4, 2.6	Issue 6
Nicholas Jackson	1.4, 2.5 (A1), 3.1, 4.1, 6, general proofreading/revision	Issue 3
Tanay Munnangi	Reviewed and edited 3.1, 3.2, 3.3	Issue 7

1 Introduction

1.1 Purpose of this document and of the system

This document is the baseline requirements specification for MidMeet. It is written for the course instructor and teaching staff, and for the project team, who will use it as the starting point for design and testing. It records what the system must do, the limits it must work within, and the questions we have not yet answered.

MidMeet helps two people who live in different places to choose a fair, real place to meet. The problem belongs to anyone who has to meet a friend or another person somewhere between two homes. Today, people guess the halfway point on a map, and the current online maps do not account for if the driving times between each person are equal. MidMeet takes each person's starting point and suggests real, named places where the two driving times or distance are as close to equal as possible.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Two participants, each with a start location	Three or more participants. This will be harder to define the midpoint, so it is a later release
Balancing travel by distance (miles) or by time (minutes)	User accounts and friend lists: not needed to meet the core goal.
Real, named places with addresses, filtered by type (e.g coffee, restaurant, park)	Booking tables or reservations needs venue integrations and not needed to meet the core goal.
A map showing both starts and the results	Walking, bicycling and public transit: each needs different routing data whose availability we have not confirmed
Driving time and distance for each participant, and the fairness gap, for each place	Live traffic and "leave at" times: depend on provider features we may not have.
A maximum driving limit per participant	

Why the exclusions were excluded:

We limited this release to driving so the team can build and test one routing method in a single semester. Groups, other travel modes, reservations and accounts each add substantial work that does not help answer the core question of where two drivers should meet.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
User	The person who operates the system, enters both addresses and picks the place
Participant	One of the two people who will meet. The user may be one of them. The other participant never uses the system
Start location	The address or position a participant will travel from
Driving miles	The road distance, in miles, a participant would drive to a place.
Driving time	The estimated minutes a participant needs to drive to a place, as returned by the routing provider
Midpoint	Two meanings. Geographic: the point halfway along a straight line between the two start locations, which ignores roads. Fair: the place where both participants' trips are closest to equal in the chosen measure. This project finds the fair midpoint and uses the geographic one only as a starting area for the search.
Place	A real, named location with a street address, not just coordinates
Geocoding	Converting an address to coordinates
Search radius	The area around the midpoint in which candidates places are looked up
Provider	The third-party service that supplies map, geocoding, routing and place data
Fairness gap	The difference between the two participants, driving times (minutes) or driving distances (miles), whichever the user chose. A smaller gap means a fairer place

1.4 References

[R1] *Google Maps Platform Documentation* https://developers.google.com/maps/documentation

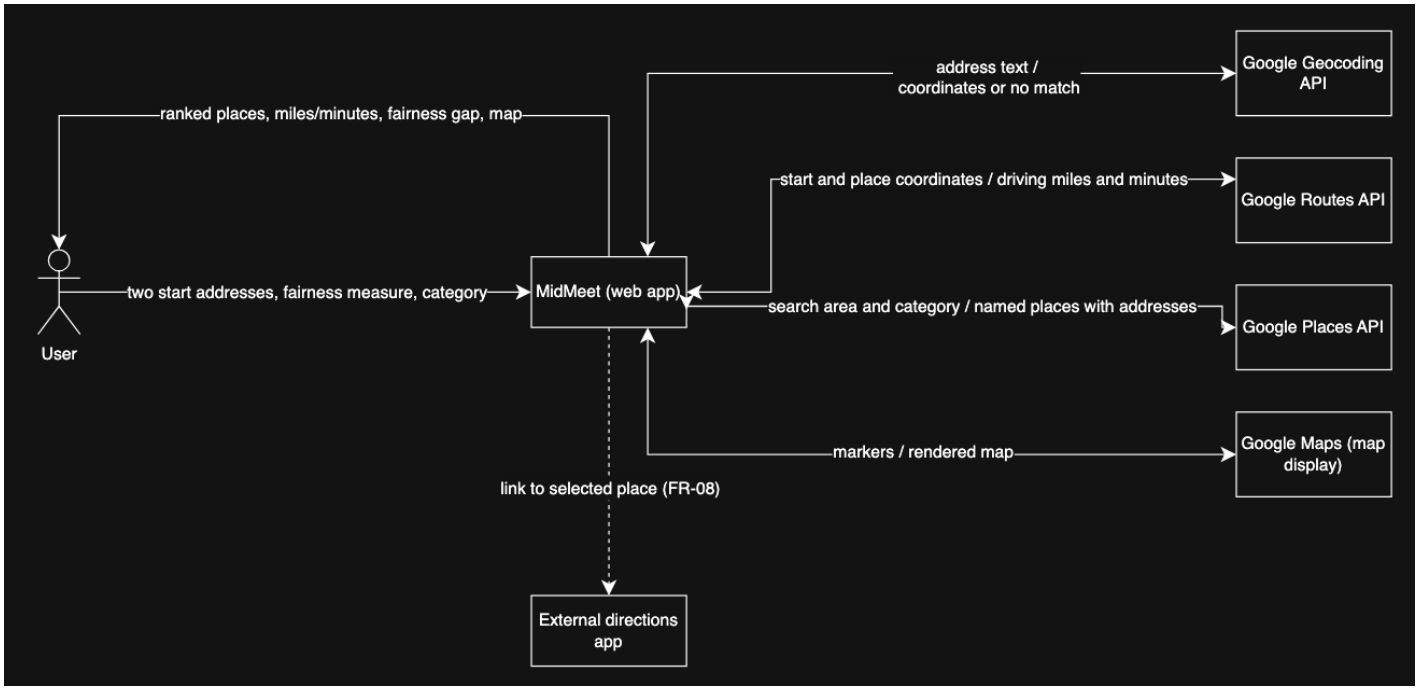
[R2] *Meeting Spaces in Dallas* https://www.visitdallas.com/blog/coworking-spaces-in-dallas/

[R3] *Three ways to add map implementation* https://mapsplatform.google.com/resources/blog/three-ways-add-map-implementation-your-app-and-when-use-each/?utm_experiment=13103221

[R4] *Student Interview* https://github.com/akao335/MidMeet/blob/main/docs/Student%20Interview

2 Overall Description

2.1 Product perspective and context



Everything inside the MidMeet box is what our team builds: taking the two start addresses and the user's choices, finding the search area, calculating each participant's driving miles and minutes and the fairness gap, ranking the places, and displaying the results. Map, geocoding, routing and place data come from Google Maps Platform (R1) and sit outside the boundary because we cannot produce this data ourselves in one semester (CON-01). The external directions app is also outside, because MidMeet only links to it (FR-09) and does not give directions itself. The other participant is not on the diagram, because they never use the system (2.2) and only see what the User shares with them. User accounts, reservations and other travel modes are out of scope (1.2), so no account or booking system appears.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
People meeting up (friends, family, partners)	Primary user; enters both addresses and picks the place	Get a fair, real place to meet in a couple of minutes, with each person's drive time shown so they can justify the choice	Assumed, based on the team's own experience arranging meetups.

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Other participant (friend, family, partner)	Never uses the system; their address is entered by the organizer	A result that is actually fair to them, and their address not stored or shared	Assumed. This stakeholder exists because the summary has one person entering both addresses (FR-01)
Course instructor and TAs	Never use the system; evaluate it and set the rules it is built under	Testable, traceable requirements; repository access; delivery by the course deadlines	P1 template and course syllabus
Project team	Builds, operates and pays for the system for the semester	Stay within free API usage; scope that seven people can finish in one semester	Team decision,
Map, routing and places data provider	Never uses the system; supplies all location data	Compliance with its terms: attribution, usage limits, limits on storing its data	Provider terms of service
Venues shown in results (cafés, restaurants, parks)	Never use the system; appear as recommendations	Correct name, address and category so they aren't misrepresented	Assumed

2.3 User characteristics

Organizer. Most organizers are students or young adults on a phone, often while messaging the other person, and want an answer within a couple of minutes. They know both people's general locations but may not know the exact address. They might type a partial address, a neighborhood, or a landmark, so the system must handle near-misses (FR-02). They can't be assumed to know map terms, and they can't be assumed to know the difference between balancing by miles and by minutes, or what a "fairness gap" is. Those choices need plain-language labels. The two participants' markers on the map must be distinguishable by more than color alone, for color-blind users.

Other participant. This person never touches the system, but they constrain it in two ways. Their address is entered by someone else without them being present, which is why start locations must not be kept (QR-02). The result also must be easy for the organizer to screenshot or share, because the other participant sees only that.

2.4 Constraints

#	Constraint	Source
C1	Map, geocoding, routing and place data must come from a third-party provider, because we cannot build this data ourselves	Team capacity in one semester; carried forward as CON-01
C2	The system must run at no cost to the team, using only providers' free usage tiers	Team has no project budget and the course provides none CON -02
C3	Use of provider data must follow the provider's terms, including attribution and limits on caching or storing results	Provider terms of service CON -03
C4	The release must be delivered by the course milestones (P2 on 8 Nov, P3 (Dec 9th)	Course syllabus
C5	Source code and project history must be kept in the team GitHub repository, with access for the instructor	Course requirement, P1 template 4.4

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Access to Routes, Geocoding, and Places API from Google or potentially a different source	The ability to produce optional meeting points, routes, and time estimations within the time constraints of the project.
A2	Google's estimated driving times are close enough to real driving times that equal estimates mean a fair meetup	A place we show as fair could give one person a much longer drive causing our main goal to fail. QR04 could pass in testing but real users could get unfair results
A3	Users enter addresses the geocoding service can find, such as a full street address, zip code, or a well-known place name.	FR 01 fails often and users keep running into the FR2 error. We would have to rely on address autocomplete, which is currently outside of our scope.
A4	At least 3 places of the chosen category exist within the starting search radius around the midpoint for most address pairs in the DFW area.	FR04 isn’t met, because the system can’t show 3-10 places. In sparse areas, searches would end up in the “too few places” case which Q2 and Q3 deals with more often than they succeed.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible, pending a spike	The core task is geocoding two addresses, searching places near the midpoint, and getting drive times from 2 origins to each candidate. All of these are standard API calls. Few of the team members have used React and are open to exploring the Google Maps API before. Spike: a script that geocodes two addresses and returns drive times to 10 nearby cafés.
Operational	Feasible	No accounts, no stored data, no staff. It runs as a web page the team deploys on a free host Vercel or Netlify. Users need only a browser and two addresses.
Schedule	Feasible with the reduced scope	The scope was cut to 2 people, driving only, and no accounts (1.2), so it fits the course's timeline. The milestones in 4.2 have named owners. Main risk: API integration taking longer than planned (see 4.3).
Economic / resource	Feasible at $0	One search uses 2 Geocoding calls, 1 Places Nearby Search, and a 2 × 20 Route Matrix (40 elements). Google Maps Platform gives 10,000 free events/month per Essentials SKU and 5,000 per Pro SKU. Route Matrix is the binding limit at about 250 searches/month, more than the ~200 needed for testing (QR-01) and demos (pricing page, retrieved 18 Oct 2026). Fallback: OpenStreetMap services (Nominatim, OSRM, Overpass) have no fees.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-01	The system shall accept a start location from each participant entered as an address.	MUST	S1, S2, R1	A tester can confirm this by entering two addresses and see both marked
FR-02	The system shall display an error message beside an invalid address field while preserving the previously entered data.	MUST	S1, R1	When a tester enters an unacceptable text as an address, the message appears next to the invalid field and the valid address is unchanged.
FR-03	The system shall let the user choose whether the fairness gap is measured in miles or minutes, defaulting to driving minutes.	MUST	S1, R1, R4	A tester runs each of 10 test address pairs twice, once selecting miles and once minutes. With miles selected, the gap is labelled in miles, and the list is sorted by ascending mile gap. With minutes selected, the gap is labelled in minutes, and the list is sorted by ascending minute gap.
FR-04	The system shall present between 3 and 10 real, named places with street addresses as candidate meeting spots.	MUST	S1, S6, R1, R2, R4	Across 10 test address pairs, every result list has 3-10 places, each with a name and address.
FR-05	The system shall let the user filter candidate places by category	MUST	S1, R1, R4	A tester selects a category, and every returned place is within that category. For example, when a tester selects "restaurant" all places in the result list are restaurants.
FR-06	The system shall display, for every candidate place, both participants' driving miles, driving minutes, and the fairness gap in the user's chosen measure.	MUST	S1, S2, R1, R4	For the top 3 results of 10 test address pairs (30 rows), each row shows two distances in miles and two times in minutes that match the routing provider's values after rounding, and a gap equal to their difference in the chosen measure, labelled with its unit.
FR-07	The system shall display both start locations and all candidate places on a map	MUST	S1, R1, R4	A tester confirms every listed place, and both starts have a marker
FR-08	The system shall display a message when no candidate meeting places are found	MUST	S1	A tester searches an area or applies a category filter with no matching places and confirms that a message appears indicating no places were found.
FR-09	The system shall provide a link from a selected place to an external directions app	COULD	S1, R1, R4	A tester clicks the link, and the directions app opens with that place.
FR-10	The system shall support three or more participants	WON'T	S4	Agreed out of this release (see 1.2) A tester confirms that the interface offers no way to enter a third address.
FR-11	The system shall support multiple transportation modes	WON'T	S4	Agreed out of this release, we will focus on car transportation for this release (see 1.2)

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
				A tester confirms that the interface offers no travel-mode option and all times and distances are for driving

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	95% of searches shall return results within 5 seconds, measured over 100 test searches on a normal mobile connection.	MUST	S1, R1, R4	A script times 100 searches and reports the 95th percentile
QR-02	Zero entered addresses shall be retained after the session ends, verified by inspecting stored data after 10 completed sessions.	MUST	S2, R1	Inspect storage after each session and inspect outgoing network requests
QR-03	At least 4 of 5 first-time testers shall complete a search, from entering two addresses to choosing a place, within 2 minutes without help	SHOULD	S1	A timed, moderated usability session with 5 testers who haven't seen the system
QR-04	For at least 18 of 20 predefined test address pairs, the top-ranked place's fairness gap shall be no more than 20% of the longer participant's driving time.	MUST	S1, S2	Compare the top result for each of the 20 pairs against the routing provider's values.

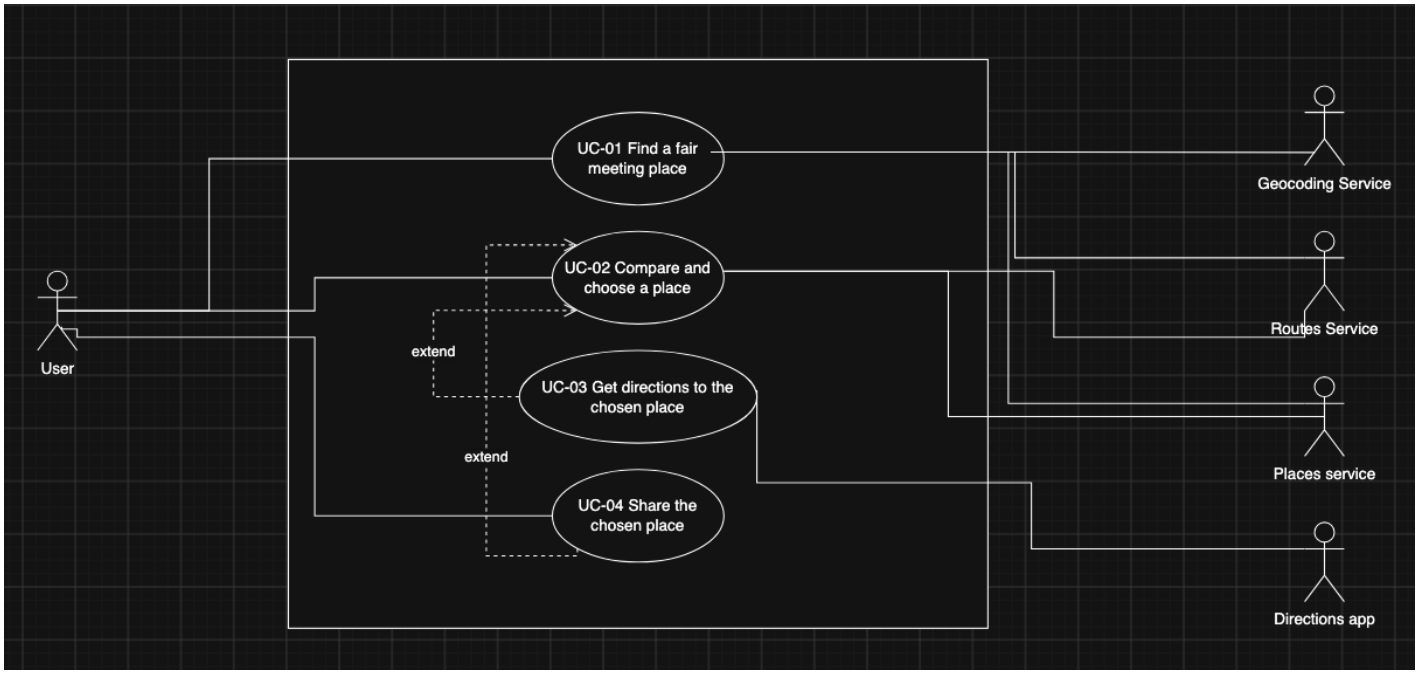
3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall obtain map, routing and place data from third-party providers	C1, R1	We cannot produce this data ourselves in one semester
CON-02	The system shall operate at no cost to the team, using only third-party providers' free usage tiers	C2, R1	The team has no project budget, and the course provides none.
CON-03	The system shall use provider data only as the provider's terms permit, including attribution and limits on caching or storing results	C3, R2	Breaking the terms can end the team's access to all location data
CON-04	The project shall be delivered by the course milestones, with all source and history kept in the team GitHub repository, accessible to the instructor	C4, C5	The course sets these dates and the repository rules.

3.4 Initial use cases

Paste the use case diagram here.

MidMeet has four use cases, each started by the User to reach a goal of their own. Entering addresses, choosing a category, and viewing the map are steps inside UC01, and not separate use cases. UC 03 and 04 extend 02, because they are optional and only happen after a place has been chosen. We wrote out UC 01 and 02 because they are the hardest to scope. UC 01 has to handle addresses that can't be found and areas with too few places, and UC 02 has to re rank the results when the users changes their category or switches between miles and minutes.



Expanded use case — copy this block for each of your two

Field	Content
Identifier and name	UC-01
Primary actor	User
Supporting actors	Geocoding service, Routes service, Places service (Google Maps Platform, R1)
Preconditions — what must be true before it starts	The User has MidMeet open in a web browser with an internet connection, and the provider services are reachable.
Trigger — what starts it	The User wants to find a place to meet that is fair for both participants.
Main success scenario — numbered steps	1. The User enters a start address for each of the two participants. 2. The User chooses whether the fairness gap is measured in minutes (default) or miles. 3. The User chooses a place category, such as coffee, restaurant or park. 4. The User asks for results. 5. The system geocodes both addresses and sets a search area around the geographic midpoint. 6. The system finds places of the chosen category in the search area. 7. The system gets each participant's driving miles and driving minutes to each place.

Field	Content
	8. The system ranks the places by smallest fairness gap and shows 3 to 10 of them. Each one shows its name, street address, both participants' miles and minutes, and the gap. 9. The system shows both start locations and all listed places on a map.
Alternative and exception flows	1a.An address cannot be found: the system shows an error next to that field, keeps the other entered data, and the User corrects it (FR-02). 1b. Both addresses resolve to the same location: the system tells the User and asks them to check the addresses. 6a. Fewer than 3 places are found: the system widens the search radius and searches again. If there are still fewer than 3, it shows a message and lets the User change the category (assumption per Q2 and Q3). 6b. A provider service fails or times out: the system shows a "try again" message and keeps everything the User entered.
Postconditions — what is true when it succeeds	The User sees 3 to 10 real, named places ranked by fairness gap, with both participants' driving miles and minutes. No start addresses are kept after the session ends (QR-02).
Requirements realized	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07,FR-08

Field	Content
Identifier and name	UC-02
Primary actor	User
Supporting actors	Routes service, Places service (R1)
Preconditions — what must be true before it starts	UC-01 has finished, and a ranked list of candidate places is on screen.
Trigger — what starts it	The User wants to compare the options and settle on one place.
Main success scenario — numbered steps	1. The User reviews the ranked list and the map. 2. The User changes the category filter, for example from coffee to park. 3. The system finds places in the new category and ranks them again by fairness gap. 4. The User switches the fairness measure between minutes and miles. 5. The system recalculates the gap for every place in the new measure, re-sorts the list, and labels the gap with its unit. 6. The User selects one place. 7. The system highlights that place on the map and shows its name, address, both participants' driving miles and minutes, and the gap.
Alternative and exception flows	3a. No places exist in the new category: the system tells the User and keeps the previous results on screen. 5a. The fairest place by minutes is not the fairest by miles: the system shows the new order. Whether it should explain the change is an open question for P2. 6a. The User wants directions: continue with UC-03 Get directions (extend). 6b. The User wants to send the result to the other participant: continue with UC-04 Share the chosen place (extend).
Postconditions — what is true when it succeeds	One place is selected, and the User can see both participants' driving miles and minutes to it.
Requirements realized	FR-03, FR-05, FR-06, FR-07, FR-08, and FR-09 (through UC-03)

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	UC-01	
FR-02	UC-01 (alternative flow 1a)	
FR-03	UC-01, UC-02	
FR-04	UC-01	
FR-05	UC-01, UC-02	
FR-06	UC-01, UC-02	
FR-07	UC-01, UC-02	
FR-08	UC-01,UC-02	
FR-09	UC-03	
FR-10		Won't for this release. No use case on purpose, because the system is driving only.

1. **A scope item with no requirement.** Section 1.2 lists "a maximum driving limit per participant" as in scope, and Q3 refers to it, but no FR or use case covers it. We propose adding an FR for it in 3.1 before P2.
2. **A use case with no requirement.** Nothing traces to UC-04 Share the chosen place, but 2.3 says the result must be easy to share because the other participant only sees what the User sends. We propose a SHOULD-priority FR for sharing the result, sourced from S2.
3. **A use case flow with no requirement.** UC-01 alternative flow 6a (fewer than 3 places, so widen the search radius) is behavior we need, but no FR covers it, so FR-04's "3 to 10 places" cannot always be met. This is logged as Q2 and Q3 in section 5, and an FR will be added once those are answered.
4. **An exception flow that belongs in 3.2.** UC-01 flow 6b (a provider service fails or times out) is not functional behavior, so we propose a quality requirement in 3.2 for how the system behaves when a provider is unavailable.
5. **Intended gaps.** FR-09 and FR-10 map to no use case on purpose, because both are WON'T for this release. Every other FR reaches at least one use case.

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Nandhana Kamalabaskaran	Organize the project plan, track deadlines, manage risks, and help coordinate team progress.	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will hold a greater responsibility for the decision they supported.
Nicholas Jackson	Determine what it will take to use Google Maps API and find no cost alternatives for geocoding, maps, and places, if necessary.	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
		hold a greater responsibility for the decision they supported.
Sameeraa Kandalgaonkar	Help with system requirements, design planning, and making sure the design matches the requirements along with Vedanshi.	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will hold a greater responsibility for the decision they supported.
Alison Kao	Draft wireframes for the main screens and checks each against the functional requirements it shows	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will hold a greater responsibility for the decision they supported.
Yair Parkhov	Creating the test cases and check the accuracy of driving times and distances to make sure the system meets its requirements.	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will hold a greater responsibility for the decision they supported.
Vedanshi Sreeramoju	Use cases, traceability and system context. Lead the P2 architecture with Sameeraa (4.2, Oct 31).	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will hold a greater responsibility for the decision they supported.
Tanay Munnangi	Review project documentation and make sure the team's work is organized and consistent.	The disagreeing sides will have the opportunity to present their side to the rest of the group, and the group will vote. This will continue until a 5-vote majority is formed. The majority voters will hold a greater responsibility for the decision they supported.

4.2 Plan to P2

Date	What will be finished	Owner
October 22nd	Review P1 requirements and if needed make necessary corrections	Nandhana Kamalabaskaram
October 25th	Research mapping APIs and confirm technical feasibility	Nicholas Jackson
October 28th	Create the basic layouts for the application's main screens	Alison Kao
October 31st	Create the basic system architecture showing how all the components will work together	Vedanshi Sreeramoju and Sameeraa Kandalgaonkar
November 4th	Prepare test cases for the main system requirements	Yair Parkhov
November 8th	Review and combine all the work for P2 and prepare for final submission	All team members

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	The mapping API can have difficulty connecting to the application	Medium	High	Nicholas will test the mapping API with the sample locations to make sure it connects properly.
R2	The application may not find enough meeting places between two locations	Medium	Medium	Vedanshi and Sameeraa will research how to expand the search area when there are not enough results.
R3	The application may give inaccurate driving times or distances	Medium	High	Yair will use test cases and compare the driving times and distances with the results from the mapping provider.
R4	Some addresses entered by the users may not be recognized by the system	Medium	Medium	Nandhana will review the address input requirements and document how the system should handle invalid addresses.
R5	The map may not display the starting locations or meeting places correctly	Medium	High	Alison will create the map layout and check that both starting locations and suggested meeting places are clearly shown on the map

4.4 Repository and tooling

https://github.com/akao335/MidMeet

All members of the team have access to the repository, including the professor. Commit messages will describe the changes made. The issues will be used to record project progress and assign tasks.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
Claude	Help structuring working for 2.1, 3.4, drafting suggestions for context diagram	Vedanshi	I drew both the diagrams myself in draw.io, I checked ever use case step and matrix row against FR-01 to FR-10, and wrote all text to match our team's decisions
Claude	Rewording the requirements in sections 3.1, 3.2, and 3.3.	Alison	Read over every requirement to confirm if my original meaning of the sentence is accurate. Checked that each requirement matches the team's scope decisions. Had other team members review as well.

Tool	What it was used for	Who used it	How the output was checked
Claude	Help drafting and refining content for 2.2, 2.3, 2.4, 2.6	Sameeraa	Read through all the content and made sure it makes sense and cross check with what I'd had planned.

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	How many meeting places should the system show if there are more than 10 possible results?	Nandhana Kamalabaskaran and Yair Parkhov	October 24th	We assume the system will display the 10 places with the smallest fairness gaps, so the users can easily compare the best options.
Q2	How large should the search radius be when looking for meeting places near the midpoint?	Alison Kao and Sameeraa Kandalgaonkar	October 30th	We assume that the system will start at a fixed search radius and then expand it if there are not enough results.
Q3	What should happen if no meeting places are found within the selected category or maximum driving limit?	Vedanshi Sreeramoju and Nicholas Jackson	November 2nd	We assume that the system will show an error message and allow the user to change their search preferences.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Nicholas Jackson		10/18/2026
Nandhana Kamalabaskaran	Nandhana Kamalabaskaran	10/18/2026
Sameeraa Kandalgaonkar		10/18/2026
Alison Kao	Alison Kao	10/18/2026
Yair Parkhov	Yair Parkhov	10/18/2026
Vedanshi Sreeramoju		10/18/2026
Tanay Munnangi	Tanay Munnangi	10/18/2026

How P1 Is Marked

P1 is worth 4% of the course grade. Following the syllabus, 80 points are for the shared team artifact and 20 for individual contribution evidence.

Criterion	Points	What full marks look like
Problem, scope and context (1.1, 1.2, 2.1)	10	The problem is a problem somebody actually has. The out-of-scope column is filled and the exclusions are justified. The context diagram's boundary matches what sections 2 and 3 claim.
Stakeholders and elicitation (2.2, 2.3)	10	Stakeholders include at least one who will never use the system. Every need has a stated basis, and assumptions are labelled as assumptions rather than disguised as findings.
Requirements quality (3.1, 3.2, 3.3)	25	Every requirement is one testable “shall” sentence with an identifier, a priority, a source and an acceptance criterion. Quality requirements carry a threshold, a condition and a method. Constraints are distinguished from quality requirements.
Use cases and traceability (3.4, 3.5)	20	Use cases are at actor-goal level, not steps. Two are written out in full, including alternative flows. The matrix is complete and the team says what the gaps mean and what they did about them.
Feasibility and plan (2.6, 4)	10	Each feasibility verdict is backed by evidence a reader can check. The plan has dates and owners. Risks are future events with actionable mitigations.
Open questions and baseline (5, 6)	5	Real open questions, with an owner and a date, and an honest statement of what was assumed in the meantime.
Document quality	—	Guidance boxes removed, sections in order, within the length budget, readable. Marks are deducted here, not awarded.
Individual contribution evidence (0.1)	20	Every member's contribution is distinguishable and traceable to something in the repository a reader can open.

▸ **WHERE MARKS ARE LOST**

Three things that cost teams the most marks every year:

1. Requirements that cannot be tested. “The system shall be user-friendly” earns nothing, however good the rest of the document is. Section 3 is 25 of the 80 team points and this is where they are lost.

2. Use cases written at step level. “Log In”, “Validate Card” and “Open Gate” are three steps of one use case. A diagram of fifteen tiny ovals scores lower than one with five real ones.

3. An empty section 5. It reads as a team that resolved its ambiguities by guessing.